

VEMULA GOPICHAND

AI Data Engineer | Cloud Data Pipelines • Databricks • Snowflake • Azure • GenAI/RAG Data Infrastructure

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SUMMARY

AI Data Engineer with 3+ years of experience building and optimizing cloud-based ETL/ELT pipelines, lakehouse architectures, and AI-ready datasets for financial services and healthcare insurance clients. Expert in Python, SQL, PySpark, Databricks, Snowflake, and Azure, with hands-on experience engineering data foundations for GenAI/RAG systems using Azure AI Search and Azure OpenAI. Proven record of cutting pipeline runtimes up to 40%, reducing cloud compute costs 15–20%, and automating data quality validation in Agile consulting environments.

TECHNICAL SKILLS

Languages: Python, SQL, PySpark, Spark SQL

Big Data & Processing: Apache Spark, Databricks, Delta Lake, ETL/ELT Pipeline Development, Batch & Incremental Processing, Data Modeling (Dimensional, Medallion Architecture), Semi-Structured Data (JSON, Parquet, CSV)

Cloud — Azure: Azure Data Factory (ADF), Azure Data Lake Storage (ADLS Gen2), Azure Databricks, Azure AI Search, Azure OpenAI, Azure DevOps

Data Warehousing: Snowflake, SQL Server, Lakehouse Architecture, Performance Tuning, Query Optimization, Compute Cost Optimization, Power BI (reporting support)

AI & ML Data Infrastructure: RAG Pipelines, Vector Embeddings, Vector Search, LLM Data Preparation, Document Chunking, Metadata Enrichment, AI-Ready Datasets

Orchestration & DevOps: Apache Airflow, Git, CI/CD, Jira, Agile/Scrum

Data Quality & Governance: Data Validation Frameworks, Source-to-Target Reconciliation, Error Handling & Logging, Schema Management, Data Documentation

PROFESSIONAL EXPERIENCE

AI Data Engineer / Consultant | Deloitte

USA | Feb 2025 – Present

- Build and maintain **20+ production ETL/ELT pipelines** processing 500 GB–1 TB daily from 10+ source systems using PySpark, Databricks, Azure Data Factory, and Airflow, feeding 60+ curated Snowflake tables that power client analytics and ML use cases.
- Reduced end-to-end pipeline runtime by **35–40%** through PySpark job tuning, partitioning redesign, and incremental load patterns — cutting selected Databricks and Snowflake compute costs by **15–20%**.
- Engineered the data foundation for an internal **GenAI knowledge-retrieval (RAG) system**: built ingestion, chunking, and metadata-enrichment pipelines converting internal documents, support tickets, and policy/process content into vector embeddings indexed in Azure AI Search and consumed by Azure OpenAI.
- Automated data quality and validation checks across critical pipelines, eliminating **6–8 hours of weekly manual validation** and increasing downstream trust in curated datasets.
- Designed dimensional and medallion-architecture (bronze/silver/gold) data models in Databricks and Snowflake, standardizing how raw and semi-structured data is curated for BI and AI/ML consumption.

Data Engineer | Hexaware Technologies

India | Jan 2021 – Oct 2023

- Developed and maintained **15–20 ETL/ELT pipelines** ingesting data from SQL Server, flat files, REST APIs, and SFTP sources into Azure Data Lake and curated Snowflake warehouse layers — supporting 40+ reporting tables for claims, member, provider, and policy analytics.
- Optimized a critical 5-hour batch pipeline down to ~3 hours (**~40% faster**) by implementing incremental load logic, tuning PySpark transformations, partitioning large datasets, and eliminating redundant joins.
- Automated source-to-target validation (row counts, nulls, duplicates, reconciliation) with Python and SQL, replacing manual Excel-based checks — saving **5–7 hours weekly** and reducing production validation errors.
- Supported migration of legacy ETL workflows to Azure Data Factory and Databricks as part of a cloud data warehouse modernization program.
- Improved pipeline reliability through standardized error handling, logging, and data quality checks; partnered with QA and business analysts to resolve data mismatches and support weekly/monthly reporting cycles, including Power BI consumers.

EDUCATION

Master of Science in Data Science | Saint Peter's University

Jersey City, NJ | May 2025